

10

Adaptations to Resistance Training

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ACTIVITY 10.2 *The Building Blocks for Increased Strength and Mass* reviews a variety of factors involved in muscle hypertrophy and how these factors interact, which can be a helpful part of developing effective training programs.

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hen he died on September 13, 2013, at the age of 84, few sport fans had heard of Jim Bradford. Bradford, an African American, spent much of his life working quietly behind the scenes at the Library of Congress as a researcher and bookbinder. At the 1952 Olympic Games in Helsinki and again at the 1960 Games in Rome, Bradford won silver medals in the weightlifting heavyweight division. Yet he was hardly known in his hometown of Washington in those decades, let alone nationally. Although it's hard to imagine in today's world of professional athletes, Bradford had to take unpaid leave from the Library of Congress to compete in the Olympics. "I come back to my job and that is it. That was par for the course then."²

Mr. Bradford was a self-proclaimed "butterball" during high school who started lifting weights after reading inspirational stories in a weightlifting magazine. He started with a set of dumbbells in his second-floor bedroom before moving his training to a nearby YMCA at his parents' request. There, he developed a unique lifting style—keeping his legs together and bending his back only as he lifted the bar overhead—for the simple reason that he feared dropping the weights, scuffing the floor, and getting kicked out of the gym!²

With any type of effective chronic exercise, multiple adaptations occur in the neuromuscular system. The type and extent of the adaptations depend on the type of training: Aerobic training, such as running, cycling, or swimming, results in little gain in muscle size and strength, but major neuromuscular adaptations occur with **resistance training**.

Resistance training was once considered inappropriate for athletes except those in competitive weightlifting, throwing events in track and field, and wrestling and boxing. Women typically avoided the weight room for fear of becoming masculine looking! But in the late 1960s and early 1970s, coaches and researchers discovered that strength and power training are beneficial for almost all sports and activities, and for women as well as men. It was not until the late 1980s and early 1990s that health professionals began to recognize the importance of resistance training to overall health, fitness, and rehabilitation.

Most athletes now include resistance training as an important component of their overall training program. Much of this attitude

change is attributable to research that has proven the performance benefits of resistance training and to innovations in training techniques and equipment. Resistance training is now an important part of the exercise prescription for all those who seek the health-related benefits of exercise.

Resistance Training and Gains in Muscular Fitness

Throughout this book, we see how important muscular fitness is to health, quality of life, and athletic performance. How do we get stronger and how do we increase muscle power and muscle endurance? Maintaining an active lifestyle is important in maintaining muscular fitness, but resistance training is necessary to increase muscular strength and power. In this section, we briefly review the changes that result from resistance training. We focus on strength, with only a brief mention of power and muscular endurance—topics that are discussed in more detail later in this book.

The neuromuscular system is one of the most responsive systems in the body to the repeated stimulation of training. Resistance training programs can produce substantial strength gains. In 3 to 6 months, one can see from 25% to 100% improvement, sometimes even more. These estimates of percentage gains in strength are, however, somewhat misleading. Most subjects in strength training research studies have never lifted weights or participated in any other form of resistance training. Most of their early gains in strength are the result of learning how to more effectively produce force and produce a true maximal movement, such as moving a barbell from the chest to a fully extended position in the bench press. This learning effect can account for as much as 50% of the early strength gains.

Muscle is very plastic, increasing in size and strength with training and decreasing in size and strength when immobilized. The remainder of this chapter details the physiological adaptations that allow people to become stronger.

Mechanisms of Gains in Muscle Strength

For many years, strength gains were assumed to result directly from increases in muscle size (hypertrophy). This assumption was logical because many people who regularly strength trained developed visually larger muscles. Also, muscles associated with a limb immobilized in a cast for weeks or months start to decrease in size (**atrophy**) and lose strength almost immediately. Gains in muscle size are generally paralleled by gains in strength, and losses in muscle size correlate highly with losses in strength. Thus, it is tempting to conclude that a direct cause-and-effect relationship exists between muscle size and muscle strength. While there is an association between size and strength, muscle strength involves far more than mere muscle size.

This does not mean that muscle size is unimportant in the ultimate strength potential of the muscle. The ability to generate force depends on the number of cross-bridges within sarcomeres, which in turn depends on the amount of actin and myosin. Size is extremely important, as revealed by the existing men's and women's world records for competitive weightlifting, shown in [figure 10.1](#). As weight classification increases (implying increased muscle mass), so does the record for the total weight lifted. However, the mechanisms associated with strength gains are more complex. What, in addition to increased size of the muscle, explains strength gains with training? Let's first consider the strong evidence that neural control of the muscle is altered with resistance training, allowing for a greater force production.

RESEARCH PERSPECTIVE 10.1

Aerobic Benefits From Resistance Exercise Training

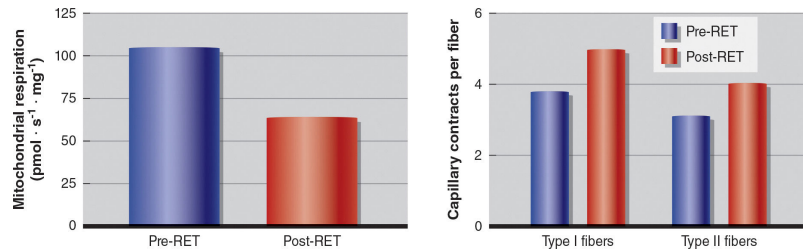
Resistance exercise training is a well-established method for increasing muscle size and strength. The classical skeletal muscle responses to resistance exercise training (e.g., hypertrophy, changes in muscle fiber type, increases in neural activation) are described in detail in this chapter. However, the long-held dogma that resistance exercise training results in distinctly different adaptations from aerobic exercise training (which are discussed in detail in [chapter 11](#)) has limited many studies to a strict focus on the mechanisms of muscle hypertrophy and gains in strength.

Conversely, increases in skeletal muscle mitochondrial number and function and increases in the number and density of capillaries in the skeletal

muscle are both well-known adaptations to aerobic exercise training. These adaptations increase ATP production and oxygen and nutrient delivery to the exercising skeletal muscle; as such, they contribute to the improvements in muscle health and maximal oxygen uptake that occur with aerobic exercise training. In addition to improving fitness, increases in mitochondrial function and skeletal muscle capillarization also enhance overall health by improving muscle bioenergetics, insulin sensitivity, and glucose tolerance at rest. Therefore, strategies to induce these adaptations may have benefits for many people, not just athletes.

Recently, exercise physiologists have begun to explore whether resistance exercise training may also exert some of these aerobic benefits. Research studies conducted at the University of Texas Medical Branch¹⁷ and Maastricht University in the Netherlands²⁶ are two of the first to explore this possibility. Both of these studies utilized a 12-week program of resistance exercise training to improve muscular fitness and collected muscle biopsy samples from the vastus lateralis before and after the training program (see figure). In Texas, the research team compared mitochondrial respiratory capacity, measured as citrate synthase activity, in the muscle biopsy samples from young subjects. They found that resistance exercise increased mitochondrial protein expression and respiratory capacity, suggesting that this resistance training protocol improved the oxidative capacity of the trained muscle. The Netherlands group examined the number of capillary contacts and the ratio of capillaries to muscle fibers in muscle biopsy samples from young and older men at baseline, and in the older men again after the resistance training protocol. (They chose to study resistance training in the older men only, opining that these subjects might have the most clinical benefit from resistance exercise to fight sarcopenia.) They found that older men had fewer capillary contacts and a lower ratio of capillaries to muscle fibers compared to young men, but, importantly, both of these aerobic-type variables increased in the older men after 12 weeks of resistance training. This led the researchers to conclude that resistance exercise can increase skeletal muscle capillarization.

Taken together, these studies show that the adaptations to resistance exercise may have more in common with aerobic exercise adaptations than we previously thought. Resistance exercise training improves mitochondrial function and capillarization in the skeletal muscle. These findings increase our understanding of how resistance exercise improves multiple fitness domains and gives us new knowledge about the health benefits of resistance training.



Twelve weeks of resistance training increases mitochondrial respiration¹⁷ and the number of capillary contacts per muscle fiber²⁶ in trained skeletal muscle. These adaptations were previously thought to occur with aerobic exercise training only. However, resistance exercise training can induce these aerobic adaptations in the trained muscle as well.

Neural Control of Strength Gains

An important component of the strength gains that result from resistance training, especially in the early stages, are neural adaptations. Enoka has made a convincing argument that strength gains can be achieved without structural changes in muscle but not without neural adaptations.⁷ Thus, strength is not solely a property of muscle but rather a property of the neuromotor system. Motor unit recruitment, frequency of motor nerve firing rates, better synchronization of motor units during a particular movement, and other neural factors are important to strength gains. Removal of neural inhibition may also play a role. These neural factors may well explain most, if not all, strength gains that occur in the absence of hypertrophy, as well as episodic superhuman feats of strength.

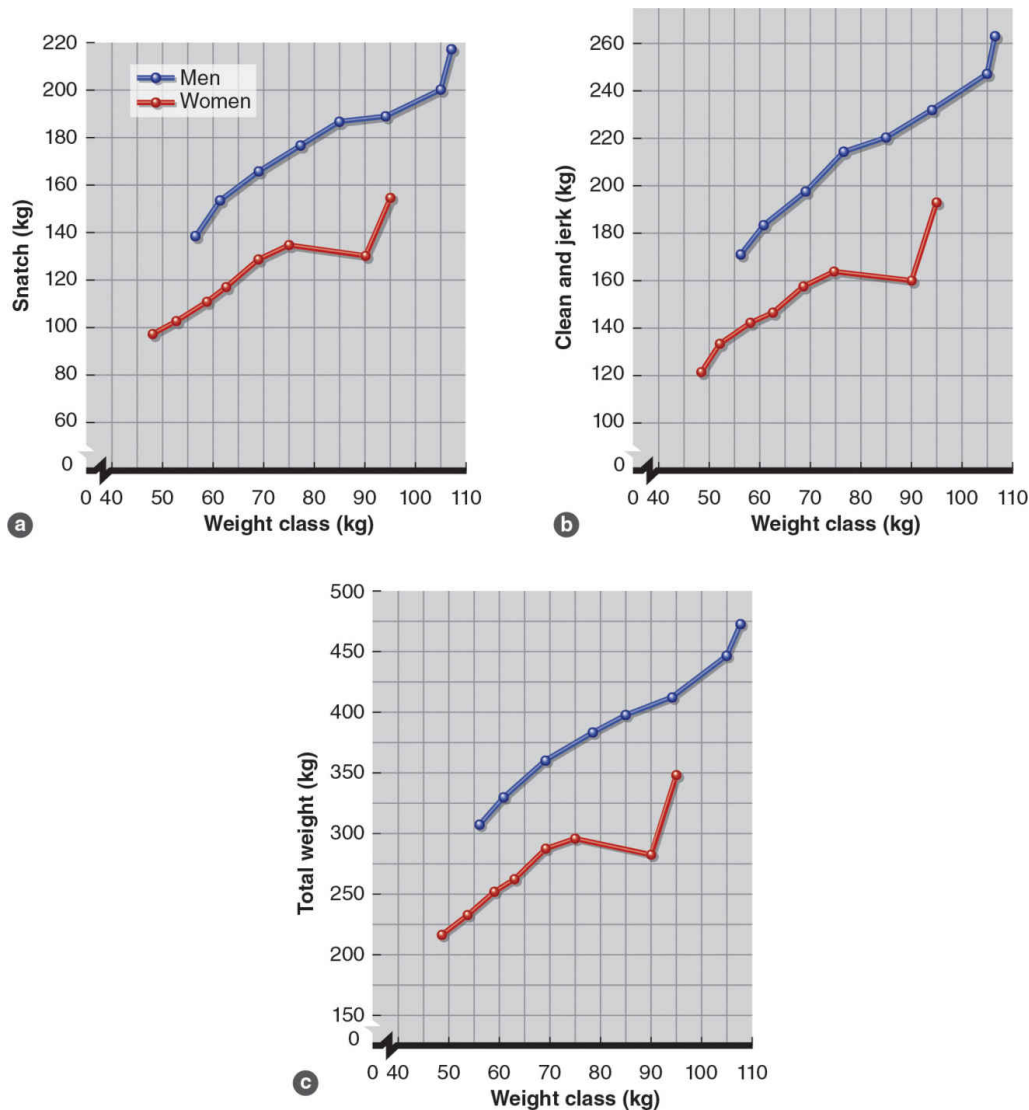


FIGURE 10.1 World records for (a) the snatch, (b) the clean and jerk, and (c) total weight for men and women through 2016.

Synchronization and Recruitment of Additional Motor Units

Motor units are generally recruited asynchronously; they are not all engaged at the same instant. They are controlled by a number of different neurons that can transmit either excitatory or inhibitory impulses (see [chapter 3](#)). Whether the muscle fibers contract or stay relaxed depends on the summation of the many impulses received by the given motor unit at any one time. The motor unit is activated and its muscle fibers contract only when the incoming excitatory impulses exceed the inhibitory impulses and the threshold is met or exceeded.

Strength gains may result from changes in the connections between motor neurons located in the spinal cord, allowing motor units to act more synchronously. This increased synchronicity means that a greater number of motor units will be firing at any one time, facilitating contraction and increasing the muscle's ability to generate force. There is good evidence to support increased motor unit synchronization with resistance training, but controversy still exists as to whether synchronization of motor unit activation produces a more forceful contraction. It is clear, however, that synchronization does improve the rate of force development and the capability to exert steady forces.⁶

Increased Rate Coding of Motor Units

The increase in neural drive of α -motor neurons could also increase the frequency of discharge, or rate coding, of their motor units. Recall from [chapter 1](#) that as the frequency of stimulation of a given motor unit increases, the muscle eventually reaches a state of tetanus, producing the absolute peak force or tension of the muscle fiber or motor unit (see [figure 1.14](#)). There is limited evidence that rate coding is increased with resistance training. Rapid movement or ballistic-type training appears to be particularly effective in stimulating increases in rate coding.

Increased Neural Drive

Neural drive refers to the combination of motor unit recruitment and rate coding of the units. Neural drive starts in the central nervous system and is spread to muscle fibers through peripheral nerves. Electromyography (EMG) using surface electrodes over the muscle measures the total activity within the nerve and muscle and therefore is a good measure of neural drive.

An alternate explanation for neutrally mediated strength gains is simply that more motor units are recruited to perform the given task, independent of whether these motor units act in unison. Such improvement in recruitment patterns could result from an increase in neural drive to the α -motor neurons during maximal contraction. Trained muscles generate a given amount of submaximal force with less EMG activity, suggesting a more efficient motor unit recruitment pattern. This increase in neural drive could increase the frequency of

discharge (rate coding) of the motor units or reduce inhibitory impulses, allowing more motor units to be activated or to be activated at a higher frequency. Additionally, maximal neural drive appears to increase with resistance training.

Autogenic Inhibition

Inhibitory mechanisms in the neuromuscular system, such as the Golgi tendon organs, might be necessary to prevent the muscles from exerting more force than the bones and connective tissues can tolerate. This control is referred to as **autogenic inhibition**. However, under extreme situations when larger forces are sometimes produced, significant damage can occur to these structures, suggesting that the protective inhibitory mechanisms can be overridden.

The function of Golgi tendon organs is discussed in [chapter 3](#). When the tension on a muscle's tendons and internal connective tissue structures exceeds the threshold of the embedded Golgi tendon organs, motor neurons to that muscle are inhibited; that is, autogenic inhibition occurs. Both the reticular formation in the brain stem and the cerebral cortex function to initiate and propagate inhibitory impulses.

Resistance training can gradually reduce or counteract these inhibitory impulses, allowing the muscle to achieve a greater force production independent of increases in muscle mass. Thus, strength gains may be achieved by reduced neurological inhibition. This theory is attractive because it can at least partially explain superhuman feats of strength and strength gains in the absence of hypertrophy.

Other Neural Factors

In addition to increasing motor unit recruitment or decreasing neurological inhibition, other neural factors can contribute to strength gains with resistance training. One of these is referred to as coactivation of agonist and antagonist muscles (the agonist muscles are the primary movers, and the antagonist muscles act to impede the agonists). If we use forearm flexor concentric contraction as an example, the biceps is the primary agonist and the triceps is the antagonist. If the two were contracting with equal force development,

no movement would occur. Thus, to maximize the force generated by an agonist, it is necessary to minimize the amount of coactivation. Reduction in coactivation could explain a portion of strength gains attributed to neural factors, but its contribution likely would be small.

Changes also have been noted in the morphology of the neuromuscular junction, with both increased and decreased activity levels that might be directly related to the muscle's force-producing capacity.

Muscle Hypertrophy

How does a muscle's size increase? Two types of hypertrophy can occur: transient and chronic. **Transient hypertrophy** is the increased muscle size that develops during and immediately following a single exercise bout. This results mainly from fluid accumulation (edema) in the interstitial and intracellular spaces of the muscle that comes from the blood plasma. Transient hypertrophy, as its name implies, lasts only for a short time. The fluid returns to the blood within hours after exercise.

Chronic hypertrophy refers to the increase in muscle size that occurs with long-term resistance training. This reflects actual structural changes in the muscle that can result from an increase in the size of existing individual muscle fibers (**fiber hypertrophy**), in the number of muscle fibers (**fiber hyperplasia**), or in both. Controversy surrounds the theories that attempt to explain the underlying cause of this phenomenon. Of importance, however, is the finding that the eccentric component of training is important in maximizing increases in muscle fiber cross-sectional area. A number of studies have shown greater hypertrophy and strength resulting solely from eccentric contraction training as compared to concentric contraction or combined eccentric and concentric contraction training. Further, higher-velocity eccentric training appears to result in greater hypertrophy and strength gains than slower-velocity training.²⁰ These greater increases appear to be related to disruptions in the sarcomere Z-lines. This disruption was originally labeled muscle damage but is now thought to represent fiber protein remodeling.²⁰ Thus, training with only concentric actions could limit muscle hypertrophy and increases in muscle strength.

Intensity and Hypertrophy

With traditional training methods, the prevailing opinion has been that an intensity of 60% to 85% of 1RM or higher is needed to achieve substantial increases in muscle size.

More recently, however, research has suggested that low-intensity exercise at <50% of 1RM can lead to gains in muscle size equal to those seen at high intensities, provided that the training is performed to volitional muscle fatigue.¹⁸ This theory holds that fatiguing contractions at light loads lead to metabolic stimuli that result in maximal muscle fiber recruitment.

Is there a minimal intensity for resistance training that will lead to muscle hypertrophy, provided that resistance exercises are performed to volitional fatigue? It has been reported that from intensities as low as 30% and as high as 90% of 1RM, load played a minimal role in stimulating muscle protein synthesis, muscle hypertrophy, and strength gains in novice exercisers.³ High-repetition (HR) and low-repetition (LR)—low and high load, respectively—training caused similar increases in skeletal muscle mass when resistance exercise was performed until volitional failure. Increases in lean body mass, as an indirect measure of muscle mass, and muscle fiber CSA, a direct measure of muscle area, occurred in both LR and HR groups with no differences between groups. There was a significant increase in 1RM strength for the leg press, knee extension, and shoulder press exercises, again with no differences between groups. These effects do not seem to depend on training status because similar results occurred in men with previous strength-training experience.¹⁶

The following section examines the two postulated mechanisms for increasing muscle size with resistance training: fiber hypertrophy and fiber hyperplasia.



Fiber Hypertrophy

Early research suggested that the number of muscle fibers in each of a person's muscles was established by birth or shortly thereafter and that this number remained fixed throughout life. If this were true, then whole-muscle hypertrophy could result only from individual muscle fiber hypertrophy. This could be explained by

- more myofibrils,
- more actin and myosin filaments,
- more sarcoplasm,
- more connective tissue, or
- any combination of these.

As seen in the micrographs in [figure 10.2](#), effective resistance training can significantly increase the cross-sectional area of muscle fibers. Such dramatic enlargement of muscle fibers does not occur, however, in all cases of muscle hypertrophy.

Muscle fiber hypertrophy is probably caused by increased numbers of myofibrils and actin and myosin filaments, which would provide more cross-bridges for force production during maximal contraction. The size of existing myofibrils does not appear to change. The increase in muscle cross-sectional area results from adding new sarcomeres in parallel to each other.

Individual muscle fiber hypertrophy from resistance training appears to result from a net increase in muscle protein synthesis. The muscle's protein content is in a continual state of flux. Protein is always being synthesized and degraded. But the rates of these processes vary with the demands placed on the body. During

exercise, protein synthesis decreases, while protein degradation increases. After exercise, although protein degradation continues, protein synthesis increases three- to fivefold more, leading to a net synthesis of myofibrillar (myosin and actin) protein. A single bout of resistance exercise can elevate net protein synthesis for up to 24 h.

Hormones and Hypertrophy

The prevailing perspective in muscle physiology is that the hormone changes induced by resistance exercise facilitate increases in muscle mass that in turn increase muscular strength. The hormones that are typically associated with this response include the anabolic hormones testosterone, growth hormone (GH), and insulin-like growth factor 1 (IGF-1). The hormone testosterone has traditionally been thought to be at least partly responsible for these changes because one of its primary functions is promoting muscle growth. Testosterone is a steroidal hormone with major anabolic functions, and men experience a significantly greater increase in muscle growth starting at puberty, which is largely due to a 10-fold increase in testosterone production. Furthermore, it has been well established that massive doses of anabolic steroids coupled with resistance training markedly increase muscle mass and strength (see [chapter 16](#)).

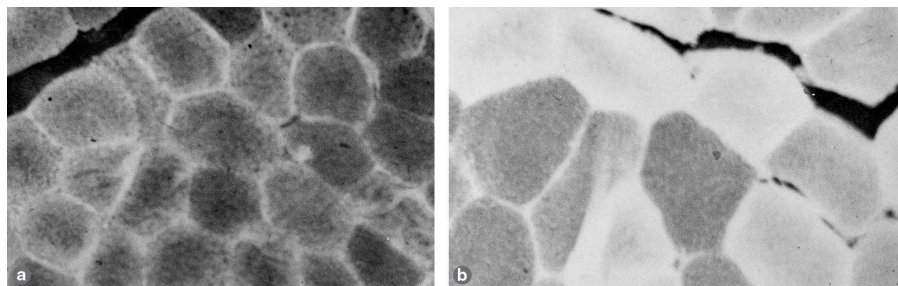


FIGURE 10.2 Microscopic views of muscle cross sections taken from the leg muscle of a man who had not trained during the previous 2 years, (a) before he resumed training and (b) after he completed 6 months of dynamic strength training. Note the significantly larger fibers (hypertrophy) after training.

While it is true that acute resistance training transiently increases the concentrations of these hormones, it has been shown experimentally that acute increases in these hormones are not required for increases in muscle mass or strength.¹⁹ Researchers at

McMaster University designed a series of studies to examine whether exercise-induced elevations in testosterone, GH, and IGF-1 were necessary for, or could enhance, muscle anabolism. They examined the elbow flexor muscles when exposed (1) to low hormone concentrations during a small muscle mass exercise consisting of isolated arm curls and (2) to high circulating hormone concentrations induced by an intense lower body exercise routine in addition to arm curls.^{29,30,31} In the low hormone trials, myofibrillar protein synthesis was elevated after acute exercise bouts, and there were gains in strength and hypertrophy after training—even though testosterone, GH, and IGF-1 all remained near baseline concentrations. This implies that postexercise increases in these hormones are not necessary to stimulate muscle anabolism. Furthermore, when these hormones were elevated postexercise, there was no further enhancement in myofibrillar protein synthesis or gain in strength and hypertrophy with training.

In addition to these studies, researchers have compared men's and women's responses to resistance training. Women have a 45-fold lower postexercise testosterone response compared to men even after their 20-fold lower baseline testosterone concentration is accounted for.²⁸ Despite having drastically lower postexercise increases in testosterone, the women were able to robustly increase rates of myofibrillar protein synthesis. Moreover, as in most exercise training studies, the men and women trained in this study had variable individual responses in terms of muscle hypertrophy. Despite these variable responses, there was no relation between each subject's exercise-induced increases in testosterone, GH, and IGF-1 and his or her muscle growth or strength gains.³²

These new data provide strong new evidence that postexercise elevations in testosterone, GH, and IGF-1 are not required to increase muscle anabolism and strength. An alternate hypothesis is that the muscle hypertrophy and strength gains occurring with resistance training are mediated by changes in intrinsic intramuscular properties.

Fiber Hyperplasia

Research on animals suggests that hyperplasia, an increase in the total number of fibers within a muscle, may also be a factor in the hypertrophy of whole muscles. Studies on cats provide fairly clear evidence that fiber splitting occurs with extremely heavy weight training.⁸ Cats were trained to move a heavy weight with a forepaw to get their food (figure 10.3). With the use of food as a powerful incentive, they learned to generate considerable force. With this intense strength training, selected muscle fibers appeared to actually split in half, and each half then increased to the size of the parent fiber.

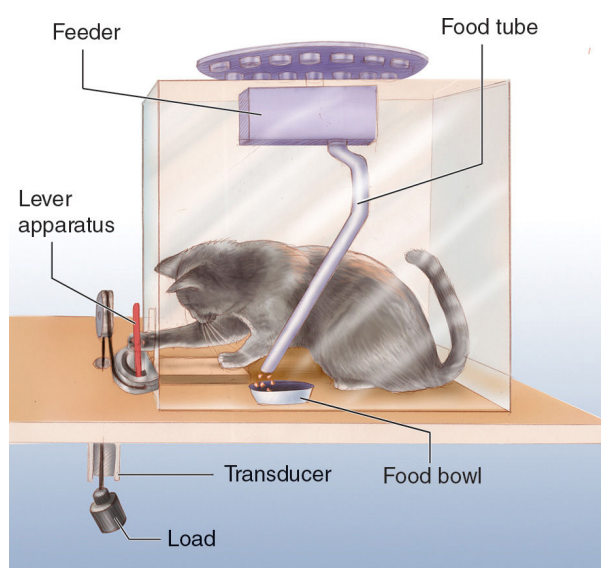


FIGURE 10.3 Heavy resistance training in cats.

Subsequent studies, however, demonstrated that hypertrophy of selected muscles in chickens, rats, and mice resulting from chronic exercise overload was attributable solely to hypertrophy of existing fibers, not hyperplasia. In these studies, each fiber in the whole muscle was actually counted. These direct fiber counts revealed no change in fiber number.

This finding led the scientists who performed the initial cat experiments to conduct an additional resistance training study with cats. This time they used actual fiber counts to determine if total muscle hypertrophy resulted from hyperplasia or fiber hypertrophy.⁹ Following a resistance training program of 101 weeks, the cats were able to perform one-leg lifts of an average of 57% of their body

weight, resulting in an 11% increase in muscle weight. Most important, the researchers found a 9% increase in the total number of muscle fibers, confirming that muscle fiber hyperplasia did occur.

The difference in results between the cat studies and those with rats and mice most likely is attributable to differences in the manner in which the animals were trained. The cats were trained with a pure form of resistance training: high resistance and low repetitions. The other animals were trained with more endurance-type activity: low resistance and high repetitions.

One additional animal model has been used to stimulate muscle hypertrophy associated with hyperplasia. Scientists have placed the anterior latissimus dorsi muscle of chickens in a state of chronic stretch by attaching weights to it, with the other wing serving as the normal control condition. In many of the studies that have used this model, the chronic stretch has resulted in substantial hypertrophy and hyperplasia.

Researchers are still uncertain about the roles played by hyperplasia and individual fiber hypertrophy in increasing *human* muscle size with resistance training. Most evidence indicates that individual fiber hypertrophy accounts for most whole-muscle hypertrophy. However, results from selected studies indicate that hyperplasia is possible in humans. It is possible that only very high intensity in resistance training can result in fiber hyperplasia, and even then, the percentage of the total muscle size increase due to this phenomenon is small, perhaps 5% to 10%. Whether strength training results in muscle fiber hyperplasia in humans remains unresolved.

In a cadaver study of seven previously healthy young men who had suffered sudden accidental death, the investigators compared cross sections of autopsied right and left tibialis anterior muscles (lower leg). Right-hand dominance is known to lead to greater hypertrophy of the left leg. In fact, the average cross-sectional area of the left muscle was 7.5% larger than that of the right. This was associated with a 10% greater number of fibers in the left muscle. There was no difference in mean fiber size.²¹

The differences between these studies might be explained by the nature of the training load or stimulus. Training at high intensities or

high resistances is thought to cause greater fiber hypertrophy, particularly of the type II (fast-twitch) fibers, than training at lower intensities or resistances.

Only one longitudinal study demonstrated the possibility of hyperplasia in men who had previous recreational resistance training experience.¹⁴ Following 12 weeks of intensified resistance training, the muscle fiber number in the biceps brachii of several of the 12 subjects appeared to increase significantly. It appears from this study that hyperplasia can occur in humans but possibly only in certain subjects or under certain training conditions.

From the preceding information, it appears that fiber hyperplasia can occur in animals and possibly in humans. How are these new cells formed? It is postulated that individual muscle fibers have the capacity to divide and split into two daughter cells, each of which can then develop into a functional muscle fiber. Importantly, satellite cells, which are the myogenic stem cells involved in skeletal muscle regeneration, are likely involved in the generation of new muscle fibers. These cells are typically activated by muscle stretching and injury; as we see later in this chapter, muscle injury results from intense training, particularly eccentric-action training. Muscle injury can lead to a cascade of responses in which satellite cells become activated and proliferate, migrate to the damaged region, and fuse to existing myofibers or combine and fuse to produce new myofibers.¹³ This is illustrated in [figure 10.4](#).

Satellite cells provide additional nuclei within muscle fibers. The added genetic machinery (DNA) is necessary to provide the increased muscle protein content and related materials to facilitate hypertrophy (and theoretically, hyperplasia).

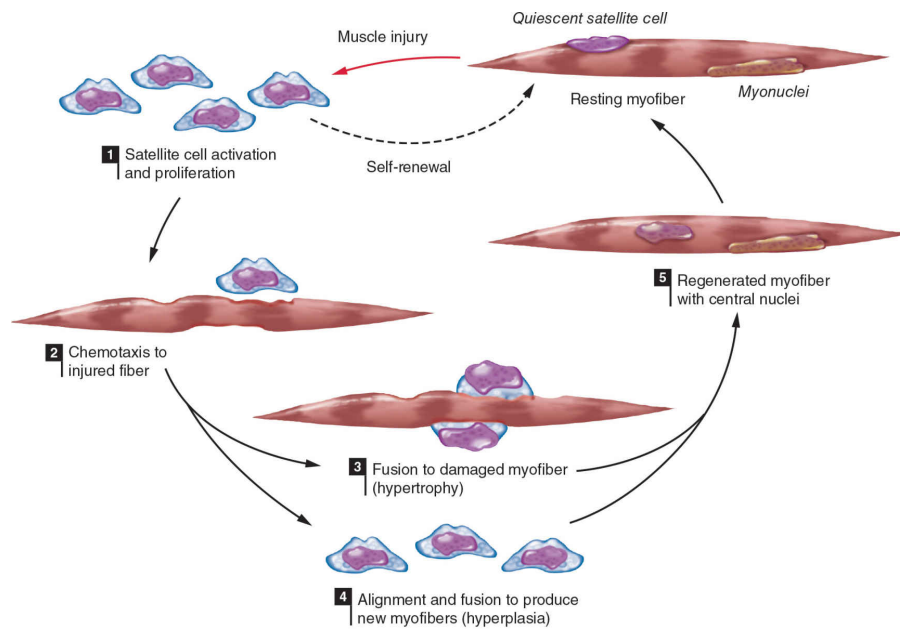


FIGURE 10.4 The satellite cell response to muscle injury.

Adapted by permission from T.J. Hawke and D.J. Garry, "Myogenic Satellite Cells: Physiology to Molecular Biology," *Journal of Applied Physiology* 91 (2001): 534-551.



Integration of Neural Activation and Fiber Hypertrophy

Research on resistance training adaptations indicates that early increases in voluntary strength, or maximal force production, are associated primarily with neural adaptations resulting in increased voluntary activation of muscle. This was clearly demonstrated in a study of both men and women who participated in an 8-week, high-intensity resistance training program, training twice per week.²² Muscle biopsies were obtained at the beginning of the study and every 2 weeks during the training period. Strength, measured according to the 1RM, increased substantially over the 8 weeks of

training, with the greatest gains coming after the second week. Muscle biopsies, however, revealed only a small, insignificant increase in muscle fiber cross-sectional area by the end of the 8 weeks of training. Thus, the strength gains were largely the result of increased neural activation.

Long-term increases in strength generally result from hypertrophy of the trained muscle. However, because it takes time to build protein through a decrease in protein degradation, an increase in protein synthesis, or both, early strength gains are typically due to changes in the pattern by which nerves activate the muscle fibers. Most research shows that neural factors contribute prominently to strength gains during the first 8 to 10 weeks of training. Hypertrophy contributes little during these initial weeks of training but progressively increases its contribution, becoming the major contributor after 10 weeks of training. However, not all studies concur with this pattern of strength development. One 6-month study of strength-trained athletes showed that neural activation explained most of the strength gains during the most intensive training months and that hypertrophy was not a major factor.¹²

Muscle Atrophy and Decreased Strength with Inactivity

When a normally active or highly trained person reduces his or her level of activity or ceases training altogether, changes occur in both muscle structure and function. This is illustrated by the results of two types of studies: studies in which entire limbs have been immobilized and studies in which highly trained people stop training—so-called detraining.

Immobilization

When a trained muscle suddenly becomes inactive through immobilization, major changes are initiated within that muscle in a matter of hours. During the first 6 h of immobilization, the rate of protein synthesis starts to decrease. This decrease likely initiates muscular atrophy, which is the wasting away or decrease in the size of muscle tissue. Atrophy results from lack of muscle use and the consequent loss of muscle protein that accompanies the inactivity. Strength decreases are most dramatic during the first week of immobilization, averaging 3% to 4% per day. This is associated not

only with the atrophy but also with decreased neuromuscular activity of the immobilized muscle.

Immobilization appears to affect both type I and type II fibers. From various studies, researchers have observed disintegrated myofibrils, streaming Z-disks (discontinuity of Z-disks and fusion of the myofibrils), and mitochondrial damage. When muscle atrophies, the cross-sectional fiber area decreases. Several studies have shown the effect to be greater in type I fibers, including a decrease in the percentage of type I fibers, thereby increasing the relative percentage of type II fibers.

Muscles can and often do recover from immobilization when activity is resumed. The recovery period is substantially longer than the period of immobilization.

Cessation of Training

Similarly, significant muscle alterations can occur when people stop training. In one study, women resistance trained for 20 weeks and then stopped training for 30 to 32 weeks. The training program focused on the lower extremity, using a full squat, leg press, and leg extension. Finally, the participants retrained for 6 weeks.²³ Strength increases were dramatic, as seen in [figure 10.5](#). Compare the women's strength after their initial training period (post-20) with their strength after detraining (pre-6). This represents the strength loss they experienced with cessation of training. During the two training periods, increases in strength were accompanied by increases in the cross-sectional²³ area of all fiber types and a decrease in the percentage of type IIx fibers. Detraining had relatively little effect on fiber cross-sectional area, although the type II fiber areas tended to decrease ([figure 10.6](#)).

To prevent losses in the strength gained through resistance training, basic maintenance programs must be established once the desired goals for strength development have been achieved. Maintenance programs are designed to provide sufficient stress to the muscles to maintain existing levels of strength while allowing a reduction in intensity, duration, or frequency of training.

In one study, men and women resistance trained with knee extensions for either 10 or 18 weeks and then spent an additional 12

weeks with either no training or reduced training.¹⁰ Knee extension strength increased 21.4% during the training period. Subjects who then stopped training lost 68% of their strength gains during the weeks they didn't train. But subjects who reduced their training (from 3 days per week to 2, or from 2 to 1) did not lose strength. Thus, it appears that strength can be maintained for at least up to 12 weeks with reduced training frequency.

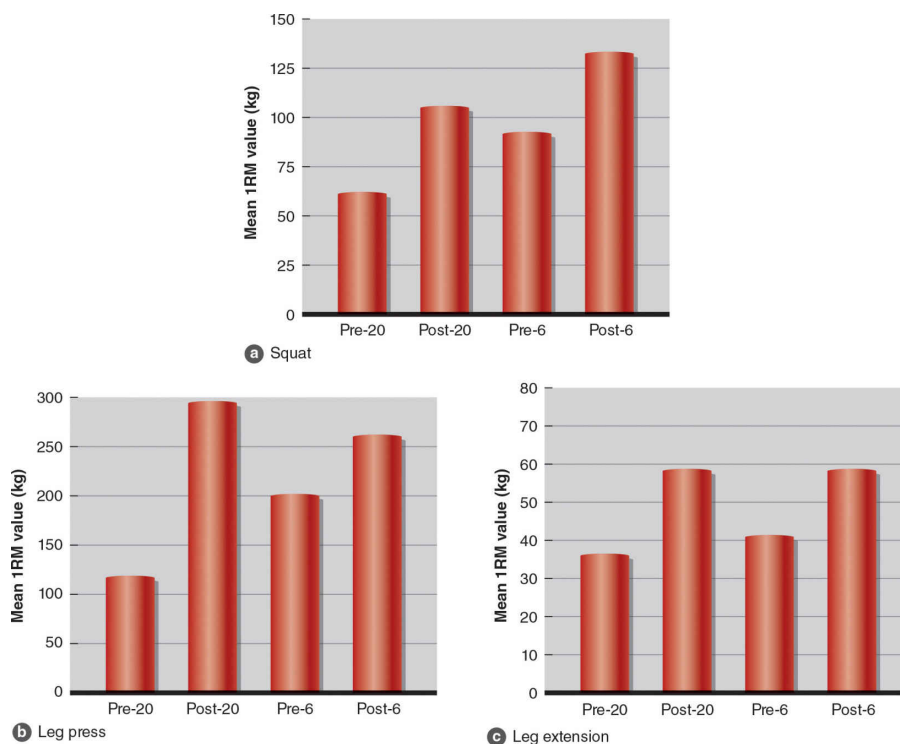


FIGURE 10.5 Changes in muscle strength for 3 different resistance exercises, (a) squat, (b) leg press, and (c) leg extension, with resistance training in women. Pre-20 values indicate strength before starting training, post-20 values indicate the changes following 20 weeks of training, pre-6 values indicate the changes following 30 to 32 weeks of detraining, and post-6 values indicate the changes following 6 weeks of retraining.

Adapted by permission from R.S. Staron et al., "Strength and Skeletal Muscle Adaptations in Heavy-Resistance-Trained Women After Detraining and Retraining," *Journal of Applied Physiology* 70 (1991): 631-640.

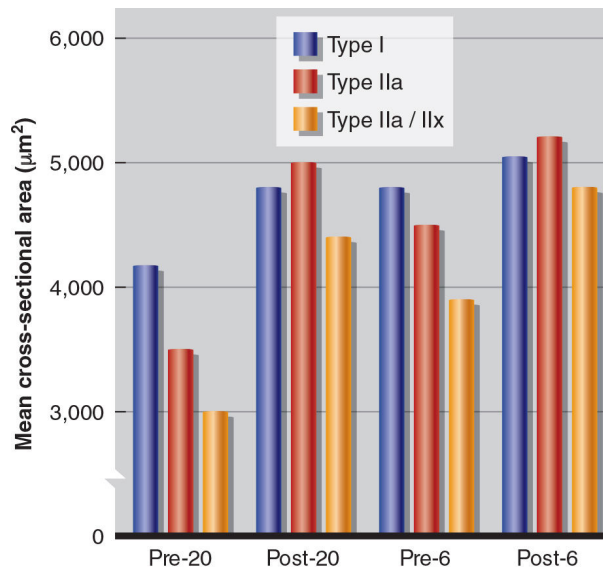


FIGURE 10.6 Changes in mean cross-sectional areas for the major fiber types with resistance training in women over periods of training (post-20), detraining (pre-6), and retraining (post-6). Type IIa/IIx is an intermediate fiber type. See [figure 10.5](#) caption for more details.

Fiber Type Alterations

Can muscle fibers change from one type to another through resistance training? The earliest research concluded that neither speed (anaerobic) nor endurance (aerobic) training could alter the basic fiber type, specifically from type I to type II or from type II to type I. These early studies did show, however, that fibers began to take on certain characteristics of the opposite fiber type if the training was of the opposite kind (e.g., type II fibers might become more oxidative with aerobic training).

Research using animal models has shown that fiber type conversion is indeed possible under conditions of cross-innervation, in which a type II motor unit is experimentally innervated by a type I motor neuron or a type I motor unit is experimentally innervated by a type II motor neuron. Also, chronic, low-frequency nerve stimulation transforms type II motor units into type I motor units within a matter of weeks. Muscle fiber types in rats have changed in response to 15 weeks of high-intensity treadmill training, resulting in an increase in type I and type IIa fibers and a decrease in type IIx fibers.¹¹ The transition of fibers from type IIx to type IIa and from type IIa to type I was confirmed by several different histochemical techniques.

Staron and coworkers found evidence of fiber type transformation in women as a result of heavy resistance training.²⁴ Substantial increases in static strength and in the cross-sectional area of all fiber types were noted following a 20-week heavy resistance training program for the lower extremity. The mean percentage of type IIx fibers decreased significantly, but the mean percentage of type IIa fibers increased. The transition of type IIx fibers to type IIa fibers with resistance training has been consistently reported in a number of subsequent studies. Further, other studies demonstrate that a combination of high-intensity resistance training and short-interval speed work can lead to a conversion of type I to type IIa fibers.

In Review

- Neural adaptations always accompany the strength gains that result from resistance training, but hypertrophy may or may not take place.
- Neural mechanisms leading to strength gains can include an increase in frequency of stimulation, or rate coding; recruitment of more motor units; more synchronous recruitment of motor units; and decreases in autogenic inhibition from the Golgi tendon organs.
- Early gains in strength appear to result more from changes in neural factors, but later long-term gains are largely the result of muscle hypertrophy.
- Transient muscle hypertrophy is the temporary enlargement of muscle resulting from edema immediately after an exercise bout.
- Chronic muscle hypertrophy occurs from repeated resistance training and reflects actual structural changes in the muscle.
- Most muscle hypertrophy results from an increase in the size of individual muscle fibers (fiber hypertrophy).
- Fiber hypertrophy increases the numbers of myofibrils and actin and myosin filaments, which provides more cross-bridges for force production.
- Muscle fiber hyperplasia has been clearly shown to occur in animal models with the use of resistance training to induce muscle hypertrophy. Only a few studies suggest evidence of hyperplasia in humans.
- Muscles atrophy (decrease in size and strength) when they become inactive, as with injury, immobilization, or cessation of training.
- Atrophy begins very quickly if training is stopped, but training can be reduced, as in a maintenance program, without resulting in atrophy or loss of strength.

- With resistance training there is a transition of type IIx to type IIa fibers.
 - Evidence indicates that one fiber type can actually be converted to the other type (e.g., type I to type II, or vice versa) as a result of cross-innervation or chronic stimulation, and possibly with training.
-

Interaction Between Resistance Training and Diet

Muscle hypertrophy in response to resistance training can be either limited or enhanced by nutrition. As mentioned previously, a net positive protein balance (more synthesis than breakdown) is the necessary condition under which muscle hypertrophy occurs. Without adequate protein in the diet, protein synthesis is compromised and muscles cannot increase their protein content and hypertrophy. Ingesting protein within a few hours after a bout of resistance exercise increases the rate of protein synthesis and thus adds to the net positive protein balance. Increased protein intake over the subsequent 24 h period will continue to support muscle anabolism. Therefore, nutrition and exercise are powerful stimulators of skeletal muscle protein synthesis.⁵

VIDEO 10.1 Presents Luc von Loon on the role of protein in adaptations to resistance training.



Recommendations for Protein Intake

An international group of researchers recently performed a systematic analysis of 49 published studies (1,863 subjects) to

determine if dietary protein supplementation enhances the gains in muscle mass and strength with resistance training.¹⁵ They surveyed randomized controlled trials in which subjects performed resistance training for at least 6 weeks and took various amounts of dietary protein supplementation. Their analysis of this large sample showed that dietary protein supplementation significantly enhanced changes in strength as measured by 1-repetition maximum tests and muscle size (fiber cross-sectional area and whole-muscle cross-sectional area). However, protein intakes greater than ~1.6 g/kg of body weight per day did not further contribute to these gains. So, although the current U.S. Dietary Reference Intake (DRI) for protein for people over 18 years of age, regardless of physical activity status, is 0.8 g/kg per day, athletes engaged in resistance training may require protein intakes in the diet as high as 1.7 g/kg per day. Although ingestion of relatively small amounts of protein (5-10 g) can stimulate muscle protein synthesis in young men and women, to make muscles larger, one should consume larger amounts of protein, on the order of 20 to 25 g, immediately after resistance exercise.¹



What type of protein should be ingested and how much? The best forms of protein for muscle hypertrophy are easily and rapidly digested and rich in essential amino acids, especially leucine. Whey protein found in milk is one source that meets both of these goals. In practice, after resistance training, athletes should consume a small amount of high-quality protein along with adequate carbohydrate in order to stimulate muscle proteins and replenish muscle glycogen stores after exercise. This can be accomplished with either a recovery beverage or foods such as milk or yogurt, a small sandwich, or a protein-rich energy bar. Adding carbohydrate to postexercise protein ingestion does not markedly affect muscle

protein balance but does have other benefits, including aiding in the resynthesis of muscle glycogen.

Is there an optimal timing of protein ingestion when an individual is trying to optimize the hypertrophic response to successive exercise sessions? A single bout of exercise stimulates muscle protein synthesis rates for several hours, and intake of protein further enhances postexercise muscle protein synthesis. The protein synthesis–stimulating effect of a single dose of amino acids is transient and lasts only 1 to 2 h. Ingesting repeated small doses of protein during recovery from resistance training may be more effective in increasing muscle hypertrophy than eating just one large meal. However, elevated muscle protein synthesis rates are not totally limited to the few hours of acute postexercise recovery. The so-called window of opportunity lasts from just before the start of resistance exercise to several hours postexercise. Providing protein before or during exercise can enhance muscle protein synthesis during exercise and is a good strategy for prolonged or repeated workouts.

Mechanism of Protein Synthesis with Resistance Training and Protein Intake

The rate of protein synthesis within the myofibrils is controlled primarily by an enzyme, or kinase, known as **mTOR** (mechanistic target of rapamycin). mTOR integrates the input from upstream pathways, including insulin and growth factors and amino acids (figure 10.7), and controls transcription of messenger RNA (mRNA). If mTOR is blocked experimentally, resistance exercise does not result in muscle hypertrophy. The primary stimulus for protein synthesis is the mechanical stretch applied to the muscle, which activates mTOR. mTOR senses cellular nutrient and oxygen levels, so it is also activated by the proper timing of protein intake, specifically proteins rich in leucine. So, delivering leucine to muscles during the window of opportunity will increase mTOR more than acute exercise alone and lead to enhanced protein synthesis and muscle hypertrophy.

The increased protein synthesis with enhanced dietary amino acid availability occurs not only because of the greater net supply of

amino acids but also because of changes in hormonal concentrations that create a more favorable anabolic environment. Insulin serves as a strong anabolic stimulus for skeletal muscle hypertrophy, as shown in [figure 10.7](#). In the presence of adequate substrate, insulin (which rises after a meal) is capable of stimulating skeletal muscle protein synthesis and hypertrophy in young muscles.

RESEARCH PERSPECTIVE 10.2

Lifting Before Bedtime for Enhanced Muscle Protein Synthesis

Ingesting protein after a bout of resistance exercise stimulates muscle protein synthesis and inhibits muscle breakdown, resulting in an overall increase in muscle protein content during the acute phase of recovery. Because of this phenomenon, postexercise protein ingestion is a widely used strategy for increasing muscle hypertrophy and speeding recovery following resistance exercise training.

Various factors can affect postexercise protein synthesis, including the amount, type, and timing of postexercise protein ingestion. Studies have shown that protein ingestion before bed increases overnight amino acid availability and stimulates muscle protein synthesis during overnight sleep. Presleep protein supplementation increases strength and hypertrophy gains over a prolonged resistance exercise training program, and protein ingestion before sleep may be a practical way to support muscle mass and maximize hypertrophy during training.

A 2016 study conducted in the Netherlands²⁵ examined whether an acute bout of resistance exercise performed in the evening could further increase the muscle protein synthesis response to presleep protein ingestion. Researchers hypothesized that by combining the powerful stimulus for protein synthesis immediately following exercise with presleep protein ingestion, they would see an even larger increase in new protein synthesis overnight. To study this, the researchers recruited 24 healthy young men and divided them into two groups: presleep protein ingestion plus evening exercise (PRO+EX) or presleep protein alone (PRO). After a standardized meal, the PRO+EX subjects performed 60 min of lower-body resistance exercise while the PRO subjects rested. After the exercise or rest session, all of the subjects consumed the same drink containing 20 g of protein. Muscle biopsy samples were taken, and a labeled amino acid isotope was continually infused for the measurement of protein turnover. Just before bed, each subject ingested another 30 g of labeled protein. Overnight, blood samples were collected at 30, 60, 90, 150, 210, 330, and 450 min while the subject slept, and a second muscle biopsy was obtained the following morning.

The results showed that overnight protein synthesis of the myofibrils was ~35% greater in the PRO+EX subjects compared to the PRO subjects. In addition, much more of the labeled dietary protein-derived amino acids were incorporated into the new myofibrils of the PRO+EX overnight. These findings led the study team to conclude that resistance exercise performed in the evening increases the overnight muscle protein synthesis response to presleep protein ingestion. Therefore, combining protein ingestion before sleep with resistance exercise may be a useful strategy when trying to maximize skeletal muscle reconditioning overnight.

Insulin stimulates protein synthesis from available amino acids by promoting a more efficient conversion of genetic codes carried by mRNA into proteins, a process known as translation. This process is accomplished by cellular organelles known as ribosomes, so it stands to reason that increasing ribosome content in the muscle fibers (that is, increasing the translational capacity) will also result in more protein being synthesized. Ribosome biogenesis, the creation of new ribosomes, appears to be an important mechanism regulating muscle size in response to resistance exercise. In fact, when the synthesis of new ribosomes is blocked biochemically, muscles fail to undergo hypertrophy. Notably, recent studies show that mTOR is involved in the synthesis of ribosomes in the cell nucleus in addition to its role in regulating translation in the cytoplasm ([figure 10.7](#)), which puts this kinase at center stage in the muscle growth process.

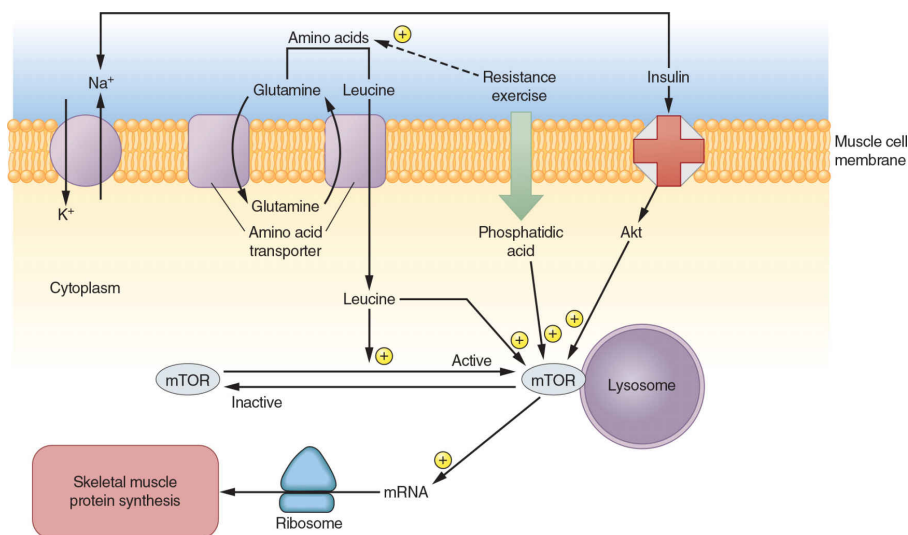


FIGURE 10.7 Schematic representation of the separate and combined roles of resistance training, insulin, and amino acid intake on skeletal muscle protein synthesis.



In Review

- Resistance exercise and protein intake are powerful stimulators of skeletal muscle protein synthesis.
- Resistance-trained athletes should consume an adequate amount of high-quality protein (as high as 1.7 g per kg of body weight per day) along with carbohydrate in order to stimulate muscle protein synthesis and also replenish muscle glycogen stores after exercise.
- The rate of protein synthesis within the myofibrils is controlled primarily by an enzyme known as mTOR. The primary stimulus for protein synthesis is the mechanical stretch applied to the muscle, which activates mTOR through a signaling pathway involving IGF-1.
- Ribosome biogenesis, the creation of new ribosomes, appears to be another important mechanism regulating muscle hypertrophy in response to resistance exercise.

Resistance Training for Special Populations

Until the 1970s, resistance training was widely regarded as appropriate only for young, healthy male athletes. This narrow concept led many people to overlook the benefits of resistance training when planning their own activities. In recent years, considerable interest has focused on training for women, children, and people who are elderly. As mentioned earlier in this chapter, the widespread use of resistance training by women, either for sport or for health-related benefits, is rather recent. Substantial knowledge has developed since the early 1970s revealing that women and men

have the same ability to develop strength but that, on average, women may not be able to achieve peak values as high as those attained by men. This difference in strength is attributable primarily to muscle size differences related to sex differences in anabolic hormones. Resistance training techniques developed for and applied to men's training seem equally appropriate for women's training. Issues of strength and resistance training for women are covered in more detail in [chapter 19](#). In this section, we first consider age, and then we summarize the importance of this form of training for all athletes, regardless of their sex, age, or sport.

Relative gains in strength appear to be similar when we compare women to men, children to adults, and elderly people to young and middle-aged adults when these gains are expressed as a percentage of their initial strength. However, the increase in the absolute weight lifted is generally greater in men compared to women, in adults compared to children, and in young adults compared to older adults. For example, after 20 weeks of resistance training, assume that a 12-year-old boy and a 25-year-old man each improves his bench press strength by 50%. If the man's initial bench press strength (1-repetition maximum, 1RM) were 50 kg (110 lb), he would have improved by 25 kg (55 lb) to a new 1RM of 75 kg (165 lb). If the boy's initial 1RM were 25 kg, he would have improved by 12.5 kg (28 lb) to a new 1RM of 37.5 kg (83 lb).

Resistance Exercise for Older Adults

Interest in resistance training for elderly people has increased since the 1980s. A substantial loss of fat-free body mass accompanies aging, a condition known as **sarcopenia**. This loss reflects mainly the loss of muscle mass, largely because most people become less active as they age. When a muscle isn't used regularly, it loses function, with predictable atrophy and loss of strength.



Can resistance training in elderly people prevent or reverse this process, and does nutrition play the same role in older people as it does in young individuals? Elderly exercisers can indeed gain strength and muscle mass in response to resistance training. This fact has important implications for both their health and quality of life (discussed in [chapter 18](#)). One of the important benefits is that, with maintained or improved strength, they are less likely to fall. This is significant because falls are a major source of injury and debilitation for elderly people and often lead to death.

Under basal conditions, fractional protein synthesis and breakdown are not much different between young and aged people. Rather, sarcopenia results from aged muscle's inability to respond appropriately to anabolic stimuli. An acute bout of resistance exercise does not appear to elicit the same hypertrophic response in skeletal muscles of older individuals. This anabolic resistance has been attributed to the inability of resistance exercise to appropriately increase mTOR signaling in the elderly.⁵ Resistance training is certainly capable of increasing strength and muscle mass in elderly persons; the response is simply blunted. With resistance training, large increases in strength are often accompanied by only small increases in myofibrillar protein and muscle size. In this age group, strength increases depend significantly on neural adaptations.

The impact of protein intake on muscle hypertrophy in the elderly is likewise blunted. Whereas as little as 5 g of protein in combination with resistance training stimulates skeletal muscle protein synthesis in young individuals, larger amounts must be ingested to cause the same effect in the elderly. This may be attributable to changes in the sensitivity of aged muscle to branched-chain amino acids. Studies indicate that ingestion of 25 to 30 g of high-quality protein or greater than 2 g of leucine is necessary to stimulate aged muscle protein synthesis to a similar degree as in young muscle. Aging is also associated with a resistance of skeletal muscle to the influence of insulin on protein synthesis, which could be a key factor in the etiology of sarcopenia.

In human aging, there is significant variation in regional body composition, and age-associated dysfunction and disability have been associated with sex and race. For example, women have greater muscle fat infiltration and higher subcutaneous fat and lower limb mass compared to men, and African Americans exhibit greater limb muscle mass that is accompanied by greater subcutaneous fat and inter- and intramuscular fat than Caucasians of the same sex.

Are there sex- and race-based differences in the physiological adaptations to resistance training in middle-aged and older adults? A study conducted at the University of Maryland used a unique one-leg strength training protocol to examine the influence of sex and race on thigh muscle volume, subcutaneous fat, and intermuscular fat

changes in response to resistance training.²⁷ Subjects, which included Caucasian and African American men and women aged 50 to 85 years, completed 10 weeks of unilateral knee extension training.

All groups had an increase in thigh muscle volume of the exercised leg. While the men experienced a greater absolute increase in muscle size, when the data were represented as a percentage increase, changes in muscle volume were similar between men and women. Nor were there any sex differences with respect to changes in subcutaneous fat or intermuscular fat. There were no differences in muscle or fat adaptations to training between Caucasian and African American exercisers. The results of this study indicate that strength training does not alter subcutaneous or intermuscular fat, regardless of sex or race. Given that there do appear to be racial differences in the incidence of metabolic disorders and functional measures of muscle quality among the elderly, other unexplored factors likely explain the racial disparity.

Resistance Training for Children

The wisdom of resistance training for children and adolescents has long been debated. The potential for injury, particularly growth plate injuries from the use of free weights, has caused much concern. Many people once believed that children would not benefit from resistance training, based on the assumption that the hormonal changes associated with puberty are necessary for gaining muscle strength and mass. We now know that children and adolescents can train safely with minimal risk of injury if appropriate safeguards are implemented. Furthermore, they can indeed gain both muscular strength and muscle mass.

Resistance training programs for children should be prescribed in much the same way as for adults, but with a special emphasis on teaching proper lifting technique. Specific guidelines have been established by a number of professional organizations, including the American Orthopaedic Society for Sports Medicine, the American Academy of Pediatrics, the American College of Sports Medicine, the National Athletic Trainers' Association, the National Strength and Conditioning Association, the President's Council on Physical

Fitness and Sports, and the U.S. Olympic Committee. Basic guidelines for the progression of resistance exercise in children are presented in [table 10.1](#).

TABLE 10.1 Basic Guidelines for Resistance Exercise Progression in Children

Age	Considerations
7 years or younger	Introduce child to basic exercises using little or no weight; develop the concept of a training session; teach exercise technique; progress from body weight calisthenics, partner exercises, and lightly resisted exercises; keep volume low.
8-10 years	Gradually increase the number of exercises; practice exercise technique in all lifts; start gradual progressive loading of exercises; keep exercises simple; gradually increase training volume; carefully monitor tolerance of the exercise stress.
11-13 years	Teach all basic exercise techniques; continue progressive loading of each exercise; emphasize exercise techniques; introduce more advanced exercises with little or no resistance. Progress to more advanced youth programs in resistance exercise; add sport-specific components; emphasize exercise techniques; increase volume.
14-15 years	Progress to more advanced youth programs in resistance exercise; add sport-specific components; emphasize exercise techniques; increase volume.
16 years or older	Move child to entry-level adult programs after all background knowledge has been mastered and a basic level of training experience has been gained.

Note. If a child of any age begins a program with no previous experience, start the child at the level for the previous age category and move him or her to more advanced levels as exercise toleration, skill, amount of training time, and understanding permit.

Reprinted by permission from W.J. Kraemer and S.J. Fleck, *Strength Training for Young Athletes*, 2nd ed. (Champaign, IL: Human Kinetics, 2005), 5.

Resistance Training for Athletes

Gaining strength, power, or muscular endurance simply for the sake of being stronger, being more powerful, or having greater muscular endurance is of relatively little importance to athletes unless it also improves their athletic performance. Resistance training by field-event athletes and competitive weightlifters makes intuitive sense. The need for resistance training by the gymnast, distance runner, baseball player, high jumper, or ballet dancer is less obvious.

RESEARCH PERSPECTIVE 10.3

Resistance Training Can Improve Health Without Changing BMI

Childhood obesity has increased dramatically over the last decade, and obese adolescents are significantly more likely to have metabolic and cardiovascular disease. Regular physical activity improves metabolic and cardiovascular health, and increasing aerobic activity is often recommended to reduce the risk of disease in obese individuals. It is well documented that aerobic training improves blood flow responses, reduces resting blood pressure, reduces inflammation, increases insulin sensitivity, and improves

body composition in overweight and obese individuals. However, adherence to aerobic training programs is very low in this unfit population. Resistance training could be an alternative strategy to improve health while increasing adherence in obese individuals. Most studies to date that have investigated resistance training to improve cardiovascular and metabolic outcomes in obesity have combined resistance training with aerobic training; because of this, little is known about the isolated effects of resistance exercise training on cardiovascular and metabolic health in obesity.

In 2015, a group of exercise physiologists in Brazil conducted a study to investigate the effects of a supervised resistance exercise training program on measures of metabolic and cardiovascular health in obese adolescents.⁴ Twenty-four obese adolescents (mean age of 14 years) performed 12 weeks of supervised resistance exercise training that included all major muscle groups. Body mass index, body composition, blood pressure, endothelial function (a measure of blood vessel health), inflammation, and insulin resistance were measured before and after training. Despite the fact that the body mass index (BMI) did not change, study participants had significantly lower body fat and waist circumferences after the 12 weeks of training. Blood pressure, endothelial function, inflammation, insulin resistance, and performance on a submaximal exercise test all improved as well.

Overall, the findings from this study led to the conclusion that resistance training improves cardiovascular and metabolic health in obese adolescents, even if BMI does not change. Although there were no changes in body mass, endothelial function, blood pressure, and metabolic profiles all improved. Resistance training programs may be an effective alternative to aerobic training to reduce the risk of cardiovascular and metabolic disease and may increase adherence to exercise programs in obese adolescents.

We do not have extensive research to document the specific benefits of resistance training for every sport or for every event within a sport. But clearly, each has basic strength, power, and muscular endurance requirements that must be met to achieve optimal performance. Training beyond these requirements, however, may be unnecessary.

Training is costly in terms of time, and athletes can't afford to waste time on activities that won't result in better athletic performances. Thus, some performance measurement is imperative to evaluate any resistance training program's efficacy. To resistance train solely to become stronger, with no associated improvement in performance, is of questionable value. However, it should also be recognized that resistance training to improve muscular endurance

can reduce the risk of injury for most sports because fatigued individuals are at an increased risk of injury.

In Review

- Resistance training can benefit almost everyone, regardless of the person's sex, age, or athletic involvement.
 - In elderly people, resistance training can slow or reverse the age-associated loss of muscle mass known as sarcopenia.
 - Aged skeletal muscle retains the ability to respond to exercise, insulin, and enhanced protein intake to substantially increase net protein synthesis. However, older muscles have a blunted response compared to young muscles.
 - Most athletes in most sports can benefit from resistance training if an appropriate program is designed for them. But to ensure that the program is working, performance should be assessed periodically and the training regimen adjusted as needed.
-

IN CLOSING

In this chapter, we have considered the role of resistance training in increasing muscular strength and improving performance. We have examined how muscle strength is gained through both muscular and neural adaptations, the importance of dietary protein intake in muscle hypertrophy, how resistance training can slow the impact of sarcopenia in the elderly, and how resistance training is of importance for both health and sport, regardless of age or sex. In the next chapter, we turn our attention away from resistance training and begin exploring how the body adapts to aerobic and anaerobic training.

KEY TERMS

atrophy
autogenic inhibition
chronic hypertrophy
fiber hyperplasia
fiber hypertrophy
mTOR
resistance training
sarcopenia
transient hypertrophy

STUDY QUESTIONS

1. What is a reasonable expectation for percentage strength gains following a 6-month resistance training program? How do these percentage gains differ by age, sex, and previous resistance training experience?
2. What is the suggested minimal intensity for resistance training that will lead to muscle hypertrophy when the exercises are performed to volitional fatigue?
3. Discuss the different theories that have attempted to explain how muscles gain strength with training.
4. What is autogenic inhibition? How might it be important to resistance training?
5. Differentiate between transient and chronic muscle hypertrophy.
6. What is fiber hyperplasia? How might it occur? How might it be related to gains in size and muscle strength with resistance training?
7. What is the physiological basis for hypertrophy?

8. Describe the respective effects of intensity and circulating hormones on muscle adaptation to fatiguing resistance training.
9. What is the physiological response to muscle immobilization?
10. To support protein synthesis during resistance training, what type of protein should be ingested and how much?
11. Is there an optimal timing of protein ingestion when an individual is trying to optimize the hypertrophic response to successive exercise sessions?
12. What is the role of mTOR in protein synthesis? How are ribosomes involved in the process?
13. How do the basic guidelines for prescribing resistance exercise for children differ from those for adults?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

[**www.HumanKinetics.com/PhysiologyOfSportAndExercise**](http://www.HumanKinetics.com/PhysiologyOfSportAndExercise)

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

